SUNWAY UNIVERSITY · SCHOOL OF MATHEMATICAL SCIENCES & COMPUTING

Autonomous Wildfire Risk Assessment & UAV Surveillance

Mamdani fuzzy inference driving a four-probe UAV inside an NVIDIA Isaac Sim digital twin

CSC3034 Computational Intelligence Supervisor: Dr. Tang Tiong Yew

NAME	STUDENT ID	ROLE
Pang Wei Sheng	24035115	Project lead — fuzzy inference engine architecture
Roshan A/L Sivakumar	23028285	Technical report — defuzzification analysis
Dinesswar A/L Sathasivam	23096910	Isaac Sim OpenUSD stage — sensor bridge
Yeoh Choon Hern	23100795	Simulation evaluation — benchmarking, A1 poster

BACKGROUND

Why a threshold is the wrong tool

THE PROBLEM

A single threshold cannot see interaction.

A 40 °C cut-off cannot separate 38 °C at 15 % humidity and 70 km/h wind from 42 °C in saturated still air. The first case is the dangerous one.

Spread is non-linear.

Dry wind at moderate temperature ignites faster than a hot, humid afternoon, so a temperature-only trigger fires late.

The telemetry is noisy.

Rotor vibration and turbulence corrupt raw readings, so a hard edge flickers between alarm states.

THE APPROACH

Continuous membership.

Four overlapping sets per variable replace the cliff edge, so the output moves smoothly with the input.

Four-probe fusion.

Temperature, relative humidity, wind speed and fuel dryness combine into one fire risk index.

The output flies the aircraft.

Throttle scales from 20 % patrol to 90 % emergency, so the UAV closes its own loop above a hotspot.

SYSTEM PIPELINE

Sensing to actuation in five stages

01

Synthetic sensing

Four probes on the OpenUSD stage, sampled at 60 Hz.

02

Fuzzification

16 input sets, four per variable, triangular and trapezoidal.

03

Fuzzy inference

10 expert rules; min for AND, max for aggregation.

04

Defuzzification

Centroid of area; continuous output at 0.88 ms on the Jetson.

05

Actuation and alert

Throttle 20-90 %; alert WATCH, WARNING or CRITICAL.

60 Hz

sensor tick from the Isaac Sim stage

0.88 ms

inference on a Jetson Orin Nano

1.2 %

of the 50 ms control budget

FUZZY ENGINE

Linguistic variables and membership functions

INPUTS

Temperature (T)

0 - 50 °C low · moderate · high · extreme

Relative humidity (RH)

0 - 100 % very low · low · moderate · high

Wind speed (WS)

0 - 100 km/h calm · moderate · high · severe

Fuel dryness (FD)

0 - 100 % wet · moderate · dry · critical

OUTPUTS

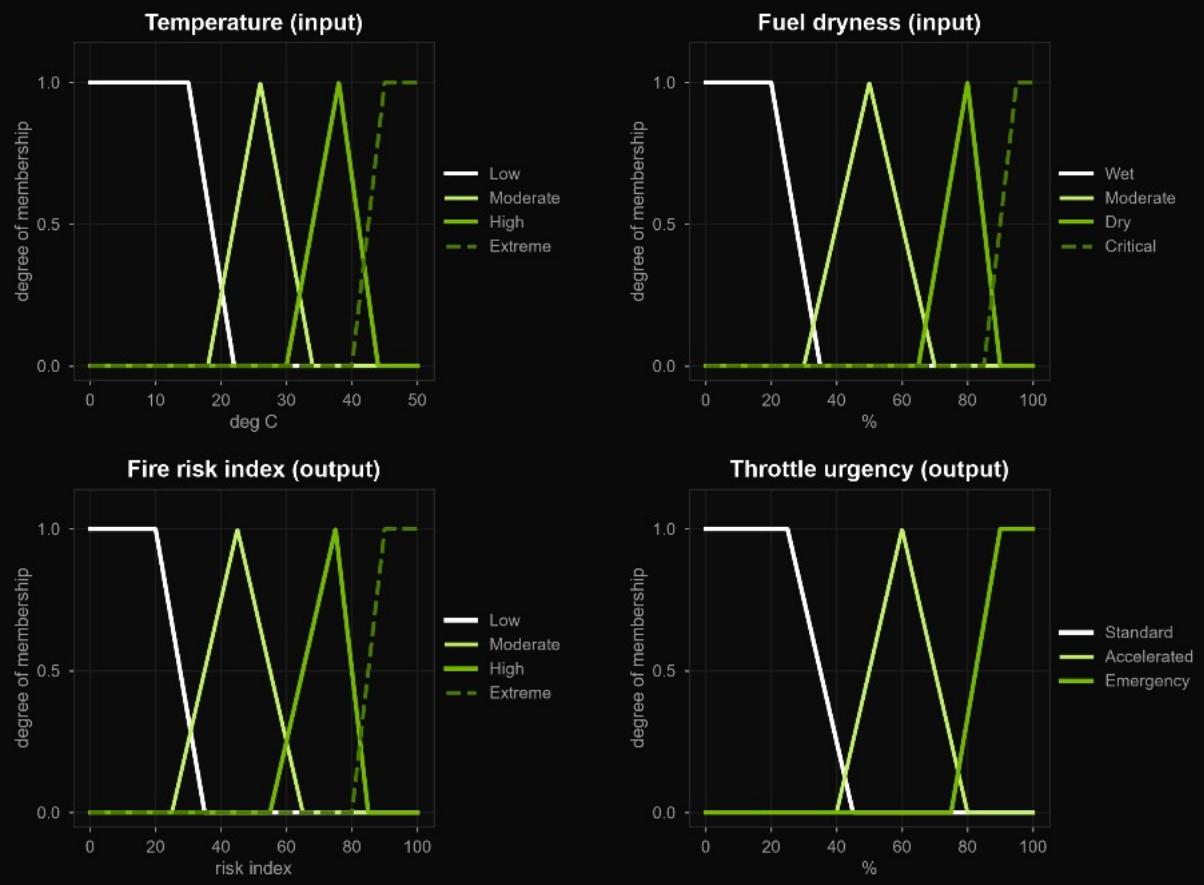
Fire risk index (FRI)

0 - 100 low · moderate · high · extreme

Throttle urgency

0 - 100 % standard · accelerated · emergency

Trapezoids close the outermost sets, so readings past the sensor range saturate instead of running off the universe.



Temperature and fuel dryness shown; humidity and wind use the same four-set template.

KNOWLEDGE BASE

Ten curated rules out of 256 combinations

RULE	ANTECEDENT	FIRE RISK	THROTTLE
R09 · R07	Temperature extreme AND (fuel critical OR humidity very low)	92	90 %
R04	Temperature moderate AND wind high AND fuel dry	71	60 %
R03	Temperature moderate AND fuel moderate	45	20 %
R01 · R10	Humidity high AND (fuel wet OR wind calm)	20	20 %

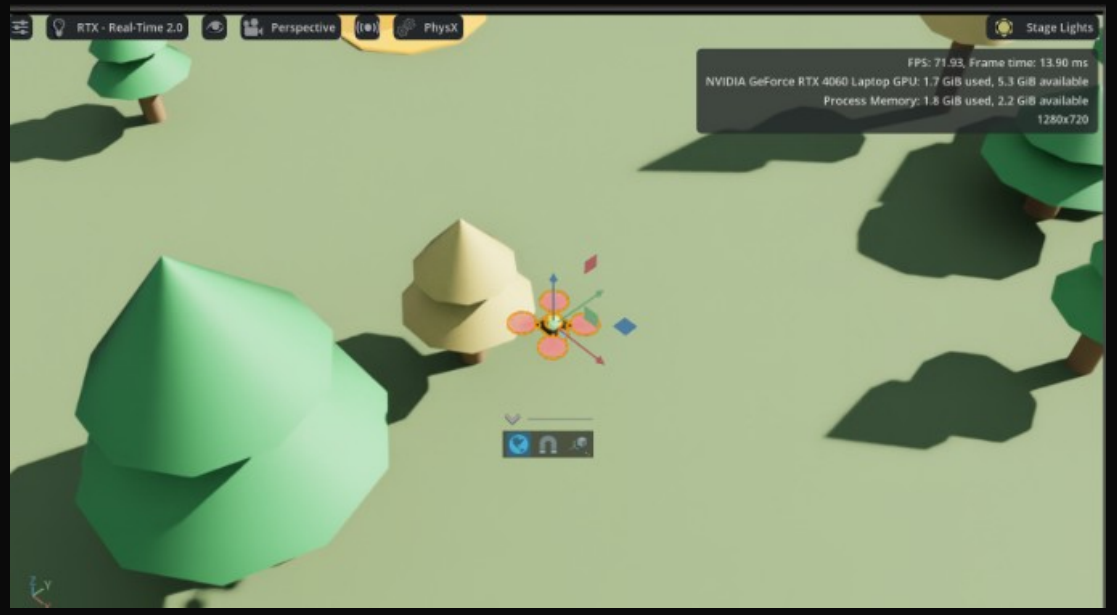
4 inputs × 4 sets = 256 antecedent combinations, curated down to the 10 that are physically meaningful.
Mamdani: min for AND, max for aggregation, centroid for defuzzification.
Cost: 0.15 ms on a workstation, 0.88 ms on the Jetson target.

3D DIGITAL TWIN

NVIDIA Isaac Sim 6.2.0 — procedural forest stage



65 m procedural forest: 48 pines, hotspot plumes, chase camera



UAV prim over hotspot 01 — probe readings feed the next inference tick

65 × 65 m

OpenUSD stage, procedurally generated

3

active hotspots, up to 44.5 °C core plumes

2

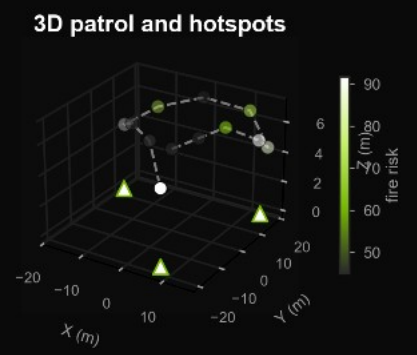
cameras: chase view and thermal gimbal

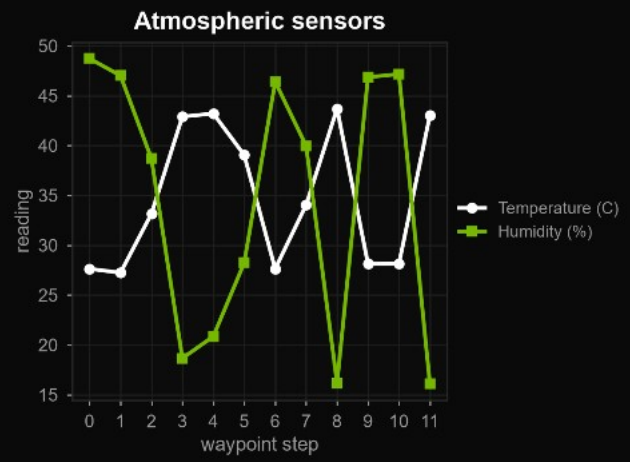
67-72

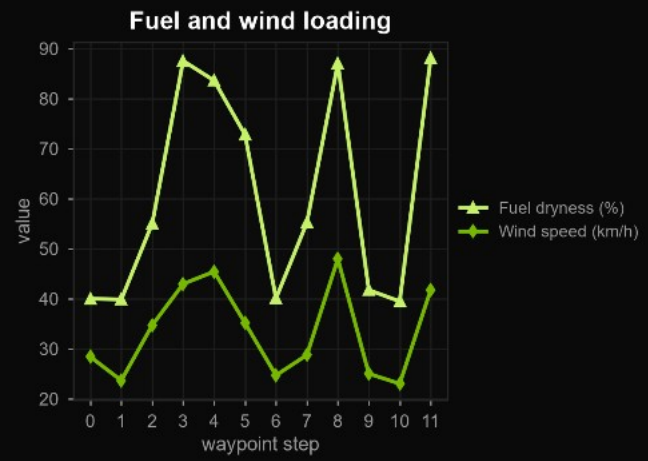
FPS in RTX Real-Time 2.0 at 1280 × 720

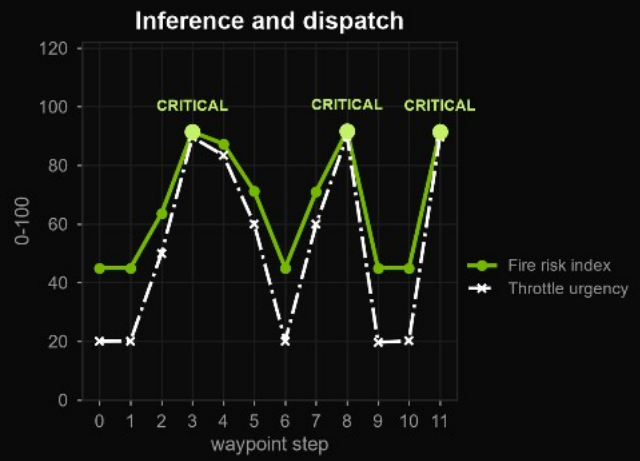
FLIGHT RESULTS

Twelve waypoints, closed loop, no retuning









WHAT THE RUN SHOWS

Baseline, WP 00-01

27.9 °C, RH 46.4 %. FRI 45.0 holds at WATCH, throttle 20 %.

Hotspot entry, WP 03

42.9 °C, RH 18.1 %, fuel 87.4 %. FRI 91.4, throttle 89.6 %.

Second plume, WP 08

43.0 °C, RH 17.9 %. FRI 91.5 — same response, no retuning.

Recovery, WP 06

28.2 °C, RH 49.2 %. FRI returns to 45.0, throttle 19.8 %.

Numbers are the printed output of the 12-waypoint simulation, reproducible from the repository.

DESIGN DECISION

Why centroid, and what it costs

Method	Avg latency	Actuator smoothness	Noise	Flight suitability
Centroid (COG)	0.884 ms	continuous	high	**selected**
Bisector	0.912 ms	good	moderate	viable
Mean of max (MOM)	0.420 ms	step jumps	low	unsafe

Continuity, not speed.

Centroid integrates the whole aggregated area, so a small input change produces a small output change. That is what keeps a 250 g airframe from seeing torque steps.

MOM jumps between peaks.

Mean of maxima returns the average of the aggregated set's maxima, so as the input crosses a boundary the output leaps to the next peak.

0.884 ms against 50 ms.

It spends 1.8 % of the control budget. MOM saves 0.46 ms and buys an unstable actuator with it.

Bisector is a compromise.

Smooth enough, but 10 % slower than centroid with no measurable benefit in the flight logs.

FEASIBILITY

It fits on the aircraft

EDGE STACK

Jetson Orin Nano, 8 GB.

7-15 W envelope, inside a micro-UAV battery budget.

0.88 ms inference.

Under 2 % of the 50 ms control loop.

Sensors already proven.

FLIR Lepton 3.5 LWIR for the plume, Sensirion SHT40 for temperature and humidity.

LoRa backhaul.

Carries the GPS fix of a confirmed hotspot to the ground crew.

WHY IT MATTERS

SDG 13.1, climate action.

Early thermal detection catches a fire before crown spread, which is where the emission saving is.

SDG 15.2, life on land.

Protects forest reserves and the habitat inside them.

Operational value.

Exact coordinates cut containment time, water use and crew exposure.

Offline by design.

The decision survives link loss, which is exactly when the fire matters most.

CONCLUSION

Delivered, and what is next

DELIVERED

Fuzzy engine.
Four inputs, ten rules, verified against the report's test vectors.

Digital twin.
65 m procedural forest holding 67-72 FPS in real-time ray tracing.

Closed-loop actuation.
20 % to 90 % throttle demonstrated over a live hotspot.

Edge speed.
0.15 ms on a workstation, 0.88 ms on the Jetson target.

NEXT

ANFIS.
Replace hand-tuned memberships with functions fitted to satellite burn history.

Multi-UAV mesh.
Decentralised coverage for areas a single airframe cannot hold.

Field trials.
PX4 and Matrice 300 under a controlled-burn permit.

Limitation, stated plainly: every result here is simulation-only, and the rule base is expert-tuned rather than learned.

Thank you

Questions, then the live Isaac Sim run

Pang Wei Sheng · Roshan A/L Sivakumar · Dinesswar A/L Sathasivam · Yeoh Choon Hern

Sunway University · CSC3034 Computational Intelligence · Dr. Tang Tiong Yew

github.com/123haiku/csc3034-fuzzy-wildfire-sim

Slides, figures and simulation output regenerate from the repository.